

# Components Notes



Center frequency  $\rightarrow f_0$

Estimate | Passband gain  $\rightarrow A_0$

Date: \_\_\_\_\_

Name: \_\_\_\_\_

Transfer function  $= \omega_0 = \sqrt{\frac{R_1 + R_2}{R_1 R_2 R_3 C_1 C_2}}$

| Particulars                                                                                                                                                                                                                    | Qty.            | Rate       | Amount |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|--------|
|                                                                                                                                                                                                                                | $R_1, R_2, R_3$ | $C_1, C_2$ |        |
| $B = \frac{1}{R_1 C_1} + \frac{1}{R_3 C_1} + \frac{1}{R_3 C_2} + \frac{1-k}{R_2 C_1}$                                                                                                                                          |                 |            |        |
| <p>① Let <math>C_1 = C_2 = 10nF</math>.</p> <p>For gain <math>\rightarrow R_1 \gg R_2 \&amp; R_3</math></p> <p>So, <math>R_1 = 6.8k\Omega</math></p> <p><math>R_2 = 9.3k\Omega</math></p> <p><math>R_3 = 2.4k\Omega</math></p> |                 |            |        |
| <p>②</p> $f_0 = \frac{1}{2\pi \times 10^{-8}} \sqrt{\frac{6800 + 1300}{6800 \times 1300 \times 2400}} = \frac{10^8}{2\pi \sqrt{8100}} = 9,825 \text{ Hz}$ <p><math>\approx 9.8 \text{ kHz}</math></p>                          |                 |            |        |
| Total                                                                                                                                                                                                                          |                 |            |        |

Amplifier gain (K) for Q & passband gain.

$$B = 10^8 \left[ \frac{1}{6800} + \frac{2}{2400} + \frac{1-K}{1300} \right]$$

$$= 174,900 - 76,900 K.$$

To achieve a good quality,  $K = 2.1$

$$K = 1 + \frac{R_f}{R_i} \quad \left\{ \begin{array}{l} R_i = 10 \text{ k}\Omega \\ R_f = 11 \text{ k}\Omega \end{array} \right.$$

$$K = 2.1$$

So,

$$B = 174,900 - 76,900(2.1)$$

$$= 13,410 \text{ rad/s.}$$

$$\text{So, } Q = \frac{\omega_0}{B} = \frac{2\pi(9825)}{13410} = 4.60.$$

satisfies  $\rightarrow Q > 2.$

④ Passband gain ( $A_0$ )

$$A_0 = \frac{K/(R_i C_1)}{B} = \frac{2.1 / (6800 \times 10^{-8})}{13410}$$

$$= 2.3 \text{ V/V.}$$

$$20 \log_{10}(2.3) = 7.23 \text{ dB.}$$